

Clothing Store Management System

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Brief on the Idea

The purpose of this database is to design a comprehensive and efficient schema for a clothing store. It is structured to manage and organize critical business operations, including product inventory, customer information, payment transactions, discounts, staff management, and branch details. The database ensures smooth operations, accurate data tracking, and enhanced reporting capabilities by defining clear relationships between these entities.

Key Features:

1. **Product Management:** Tracks details about available products, including name, description, price, stock levels, categories, and sizes.
2. **Customer Management:** Manages customer profiles, including their contact details, shipping addresses, and membership status.
3. **Payment Processing:** Records payment details such as the amount, date, method, and transaction status.
4. **Discount Allocation:** Defines applicable discounts, their percentages, and valid timeframes, ensuring flexible promotional campaigns.
5. **Staff Administration:** This department tracks staff details, including their assignments to specific branches and their roles within the company.
6. **Branch Operations:** Manages branch-specific data such as location, contact information, and associated staff.

SQL Code

1. Creating the Schema

```
1 CREATE SCHEMA Clothing_Store;
```

This is the command to create a schema named Clothing_Store in a database for the Clothing Store Management System.

2. Creating the Tables

- Products Table

```
1 CREATE TABLE Products (  
2     prod_id int PRIMARY KEY,  
3     name VARCHAR(100),  
4     price DECIMAL(10, 2),  
5     category VARCHAR(50),  
6     sizes VARCHAR(20)  
7 );
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0011 seconds.)

```
SELECT * FROM `Products`
```

prod_id	name	price	category	sizes
---------	------	-------	----------	-------

The Products table in SQL defines attributes for a product inventory:

- prod_id: A unique identifier for each product (Primary Key).
- name: The product's name is limited to 100 characters.
- price: A decimal value representing the product's price with up to 10 digits, including 2 decimal places.
- category: Specifies the product category, such as shirts, pants, or accessories, limited to 50 characters.
- sizes: Indicates the available size(s) of the product (e.g., S, M, L), limited to 20 characters.

- Customers

```
1 CREATE TABLE Customers (  
2     customer_id int PRIMARY KEY,  
3     F_name VARCHAR(50),  
4     Mid_name VARCHAR(50),  
5     L_name VARCHAR(50),  
6     phone VARCHAR(15),  
7     email VARCHAR(100),  
8     address VARCHAR(255),  
9     loyalty_points int  
10 );  
11
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0012 seconds.)

```
SELECT * FROM `Customers`
```

customer_id	F_name	Mid_name	L_name	phone	email	address	loyalty_points
-------------	--------	----------	--------	-------	-------	---------	----------------

The Customers table in SQL defines attributes for customer information:

- customer_id: A unique identifier for each customer (Primary Key).
- F_name: The customer's first name, limited to 50 characters.
- Mid_name: The customer's middle name, limited to 50 characters.
- L_name: The customer's last name, limited to 50 characters.
- phone: The customer's phone number, is limited to 15 characters.
- email: The customer's email address, is limited to 100 characters.
- address: The customer's address, is limited to 255 characters.
- loyalty_points: An integer representing the customer's loyalty points.

- Payments Table

```
1 CREATE TABLE Payment (  
2     payment_id int PRIMARY KEY,  
3     amount DECIMAL(10, 2),  
4     payment_date DATE,  
5     payment_method VARCHAR(50),  
6     customer_id int,  
7     staff_id int,  
8     prod_id int,  
9     quantity int NOT NULL,  
10    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),  
11    FOREIGN KEY (prod_id) REFERENCES Products(prod_id),  
12    FOREIGN KEY (staff_id) REFERENCES Staff(staff_id)  
13 );|
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
SELECT * FROM `Payment`  
.....
```

payment_id	amount	payment_date	payment_method	customer_id	staff_id	prod_id	quantity
------------	--------	--------------	----------------	-------------	----------	---------	----------

The Payment table in SQL defines attributes for managing payment transactions:

- payment_id: A unique identifier for each payment (Primary Key).
- amount: The total payment amount is stored as a decimal value with up to 10 digits and 2 decimal places.
- payment_date: The date when the payment was made.
- payment_method: Specifies the payment method (e.g., cash, credit card), limited to 50 characters.
- customer_id: A foreign key referencing the Customers table (customer_id).
- staff_id: A foreign key referencing the Staff table (staff_id).
- prod_id: A foreign key referencing the Products table (prod_id).
- quantity: The number of product units purchased; it cannot be null.

The table also enforces referential integrity through foreign keys to link Customers, Products, and Staff tables.

- Branches Table

```
1 CREATE TABLE Branches (  
2     branch_id int PRIMARY KEY,  
3     location VARCHAR(255),  
4     working_hours VARCHAR(50),  
5     opening_date DATE  
6 );|
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
SELECT * FROM `Branches`
```

branch_id	location	working_hours	opening_date
-----------	----------	---------------	--------------

The Branches table defines attributes for each store's branch information.

- branch_id: A unique identifier for each branch (Primary Key).
- location: The branch's address, is limited to 255 characters.
- working_hours: The branch's working hours, are limited to 50 characters.
- opening_date: The date when the branch was opened.

- Discounts Table

```
1 CREATE TABLE Discounts (  
2     discount_id int PRIMARY KEY,  
3     description VARCHAR(255),  
4     start_date DATE,  
5     end_date DATE,  
6     percentage DECIMAL(5, 2)  
7 );
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
SELECT * FROM `Discounts`
```

discount_id	description	start_date	end_date	percentage
-------------	-------------	------------	----------	------------

This table stores all the store's discount codes and details which include:

- discount_id: A unique identifier for each discount (Primary Key).
- description: A textual description of the discount, limited to 255 characters.
- start_date: The date when the discount starts.
- end_date: The date when the discount ends.
- percentage: The discount percentage, stored as a decimal value with up to 5 digits and 2 decimal places (e.g., 25.50)

- Staff Table

```

1 CREATE TABLE Staff (
2     staff_id INT PRIMARY KEY,
3     F_name VARCHAR(50),
4     Mid_name VARCHAR(50),
5     L_name VARCHAR(50),
6     phone VARCHAR(15),
7     email VARCHAR(100),
8     role VARCHAR(50),
9     branch_id int NOT NULL,
10    hire_date DATE,
11    salary DECIMAL(10, 2),
12    FOREIGN KEY (branch_id) REFERENCES Branches(branch_id)
13 );

```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0013 seconds.)

SELECT * FROM `Staff`

staff_id	F_name	Mid_name	L_name	phone	email	role	branch_id	hire_date	salary
----------	--------	----------	--------	-------	-------	------	-----------	-----------	--------

The Staff table in SQL stores information about the employees, including their details, roles, salary, and the branch they are assigned to.

- staff_id: A unique identifier for each staff member (Primary Key).
- F_name: The staff member's first name, limited to 50 characters.
- Mid_name: The staff member's middle name, limited to 50 characters.
- L_name: The staff member's last name, limited to 50 characters.
- phone: The staff member's phone number, is limited to 15 characters.
- email: The staff member's email address, is limited to 100 characters.
- role: The role of the staff member (e.g., manager, teller), is limited to 50 characters.
- branch_id: A foreign key referencing the Branches table (branch_id) to associate the staff member with a specific branch.
- hire_date: The date when the staff member was hired.
- salary: The staff member's salary, stored as a decimal value with up to 10 digits and 2 decimal places (e.g., 50000.00)

3. Relationships

Most relationships were included in the tables such as:

- Sells
- Works_at
- Pay
- Purchased
- Processes

But others such as *USES*, *STOCKS*, and *PROCESSES* are separate tables in the SQL.

USES

```
1 CREATE TABLE Uses (  
2     customer_id INT,  
3     discount_id INT,  
4     PRIMARY KEY (customer_id, discount_id),  
5     FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),  
6     FOREIGN KEY (discount_id) REFERENCES Discounts(discount_id)  
7 );
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
SELECT * FROM `Uses`
```

customer_id	discount_id
-------------	-------------

This table keeps track of the relationship between Customers using Discounts.

It was a separate as it is a Many-Many relationships.

STOCKS

```
1 CREATE TABLE Stocks (  
2     prod_id int,  
3     branch_id int,  
4     quantity int,  
5     PRIMARY KEY (prod_id, branch_id),  
6     FOREIGN KEY (prod_id) REFERENCES Products(prod_id),  
7     FOREIGN KEY (branch_id) REFERENCES Branches(branch_id)  
8 );
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
SELECT * FROM `Stocks`
```

prod_id	branch_id	quantity
---------	-----------	----------

This table keeps track of the relationship between Products in-stock in Branches.

The reason for it being in a separate table is also being Many-Many relationships with an additional attribute: quantity.

4. Inserting

Customers

```
1 INSERT INTO Customers VALUES
2 (234, "Selena", "V", "Gonzalez", 01007894561, "selenagon@gmail.com", "445 Fredricks , Chicago" , 30),
3 (566, "Winston", "L", "Schmidt", 0112345678, "schmidt@gmail.com", "23 JapAvenue , Michigan" , 24),
4 (334, "Tamar", "K", "Jabar", 0156789900, "tamy@gmail.com", "44 Marwa Dst, Jeddah" , 39),
5 (1, "Maya", "F", "Karter", 0124467890, "maKarter4@gmail.com", "75 Cornelia St, New York", 100),
6 (23, "Lewis", "H", "Marigold", 012345678901, "leWis@gmail.com", "45 Fawzi Moaz, Smouha", 78);|
```

Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

SELECT * FROM Customers

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

	customer_id	F_name	Mid_name	L_name	phone	email	address	loyalty_points
☐ Edit Copy Delete	1	Maya	F	Karter	124467890	maKarter4@gmail.com	75 Cornelia St, New York	100
☐ Edit Copy Delete	23	Lewis	H	Marigold	12345678901	leWis@gmail.com	45 Fawzi Moaz, Smouha	78
☐ Edit Copy Delete	234	Selena	V	Gonzalez	1007894561	selenagon@gmail.com	445 Fredricks , Chicago	30
☐ Edit Copy Delete	334	Tamar	K	Jabar	156789900	tamy@gmail.com	44 Marwa Dst, Jeddah	39
☐ Edit Copy Delete	566	Winston	L	Schmidt	112345678	schmidt@gmail.com	23 JapAvenue , Michigan	24

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Branches

```
1 INSERT INTO Branches VALUES
2 (1, "55 GrandAvenue, Alex" , "14hrs/day", "2002-02-20"),
3 (2, "34 Bayland , Riyadh" , "10hrs/day" , "2010-10-10"),
4 (3, "99 WillowSt , New York" , "9hrs/day" , "2012-11-29"),
5 (4, "45 GHive Dst , Colorado" , "9hrs/day" , "2013-01-04"),
6 (5, "66 St.Mary Av, Rome" , "6hrs/day" , "2017-04-29");|
```

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

	branch_id	location	working_hours	opening_date
☐ Edit Copy Delete	1	55 GrandAvenue, Alex	14hrs/day	2002-02-20
☐ Edit Copy Delete	2	34 Bayland , Riyadh	10hrs/day	2010-10-10
☐ Edit Copy Delete	3	99 WillowSt , New York	9hrs/day	2012-11-29
☐ Edit Copy Delete	4	45 GHive Dst , Colorado	9hrs/day	2013-01-04
☐ Edit Copy Delete	5	66 St.Mary Av, Rome	6hrs/day	2017-04-29

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Products

```
1 INSERT INTO Products VALUES
2 (231000000 , "TrZ grey sweater" , 39.99,"UNISEX","XS-XL"),
3 (231000469 , "vrM black coat" , 78.75 , "Women" , "S-XL"),
4 (231003562 , "JFK-Y orange jeans" , 101.84 , "Men" , "L-XL"),
5 (231007758 , "beY olivegreen jacket" , 45.45 , "Men" , "S-XL"),
6 (236767676, "YZMY blue hoodie", 67.67 , "Children" , "4-8 years");|
```

+ Options

				prod_id	name	price	category	sizes
	Edit	Copy	Delete	231000000	TrZ grey sweater	39.99	UNISEX	XS-XL
	Edit	Copy	Delete	231000469	vrM black coat	78.75	Women	S-XL
	Edit	Copy	Delete	231003562	JFK-Y orange jeans	101.84	Men	L-XL
	Edit	Copy	Delete	231007758	beY olivegreen jacket	45.45	Men	S-XL
	Edit	Copy	Delete	236767676	YZMY blue hoodie	67.67	Children	4-8 years

☐ Check all
 With selected:
 ☐ Edit
 ☐ Copy
 ☐ Delete
 ☐ Export

☐ Show all
 Number of rows: 25
 Filter rows: Search this table
 Sort by key: None

Discounts

```
1 INSERT INTO Discounts VALUES
2 (50 , "STOREBDAY" , "2025-02-20" , "2025-03-20" , 50.00),
3 (10, "Christmas" , "2024-12-01","2025-01-05" , 10.00),
4 (20,"Ramadan" , "2025-03-01" , "2025-04-01",20.00),
5 (75,"NewOpenning" , "2025-09-08" , "2025-10-08" , 75.00),
6 (30,"paRisFasHionWeek" , "2025-01-17","2025-01-21",30.00);
```

☐ Show all
 Number of rows: 25
 Filter rows: Search this table
 Sort by key: None

+ Options

				discount_id	description	start_date	end_date	percentage
☐	Edit	Copy	Delete	10	Christmas	2024-12-01	2025-01-05	10.00
☐	Edit	Copy	Delete	20	Ramadan	2025-03-01	2025-04-01	20.00
☐	Edit	Copy	Delete	30	paRisFasHionWeek	2025-01-17	2025-01-21	30.00
☐	Edit	Copy	Delete	50	STOREBDAY	2025-02-20	2025-03-20	50.00
☐	Edit	Copy	Delete	75	NewOpenning	2025-09-08	2025-10-08	75.00

☐ Check all
 With selected:
 ☐ Edit
 ☐ Copy
 ☐ Delete
 ☐ Export

☐ Show all
 Number of rows: 25
 Filter rows: Search this table
 Sort by key: None

Staff

```

1 INSERT INTO Staff VALUES
2 (1,"Jessica","F","Day","0987786752","jessday@hotmail.com","Assistant Manager", 3 ,"2013-01-09",8500),
3 (2,"Alessandro","B","Newt","890764563","ales@hotmail.com","Stock Manger",2,"2011-08-08",4900),
4 (3,"Farida","M","Tamer", "56784453344","frda@hotmail.com","Website Developer",1,"2019-05-23",6300),
5 (4,"Mohamed","A","Aly","67893356725","moh@hotmail.com","Regional Manager",4,"2020-02-29",7200),
6 (5,"Nick","J","Miller","66265430099","nmiller@hotmail.com","Finance",5,"2023-07-01",9000);

```

Options	staff_id	F_name	Mid_name	L_name	phone	email	role	branch_id	hire_date	salary
☐ Edit ☐ Copy ☐ Delete	1	Jessica	F	Day	0987786752	jessday@hotmail.com	Assistant Manager	3	2013-01-09	8500.00
☐ Edit ☐ Copy ☐ Delete	2	Alessandro	B	Newt	890764563	ales@hotmail.com	Stock Manger	2	2011-08-08	4900.00
☐ Edit ☐ Copy ☐ Delete	3	Farida	M	Tamer	56784453344	frda@hotmail.com	Website Developer	1	2019-05-23	6300.00
☐ Edit ☐ Copy ☐ Delete	4	Mohamed	A	Aly	67893356725	moh@hotmail.com	Regional Manager	4	2020-02-29	7200.00
☐ Edit ☐ Copy ☐ Delete	5	Nick	J	Miller	66265430099	nmiller@hotmail.com	Finance	5	2023-07-01	9000.00

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Payments

```

1 INSERT INTO Payment VALUES
2 (100, 199.99, '2023-11-25', 'Credit Card', 234, 1, 231000000, 2),
3 (200, 49.99, '2023-12-01', 'Cash', 566, 2, 231000469, 1),
4 (300, 129.95, '2023-12-10', 'Debit Card', 334, 3, 236767676, 3),
5 (400, 24.99, '2023-12-15', 'Online', 1, 3, 231003562, 1),
6 (500, 99.90, '2023-12-20', 'Mobile Payment', 23, 5, 231007758, 2);

```

Options	payment_id	amount	payment_date	payment_method	customer_id	staff_id	prod_id	quantity
☐ Edit ☐ Copy ☐ Delete	100	199.99	2023-11-25	Credit Card	234	1	231000000	2
☐ Edit ☐ Copy ☐ Delete	200	49.99	2023-12-01	Cash	566	2	231000469	1
☐ Edit ☐ Copy ☐ Delete	300	129.95	2023-12-10	Debit Card	334	3	236767676	3
☐ Edit ☐ Copy ☐ Delete	400	24.99	2023-12-15	Online	1	3	231003562	1
☐ Edit ☐ Copy ☐ Delete	500	99.90	2023-12-20	Mobile Payment	23	5	231007758	2

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

- Stocks

```

1 INSERT INTO Stocks VALUES
2 (231000000,1,800),
3 (231000469,2,567),
4 (231000000,4,789),
5 (236767676,5,444),
6 (231003562,3,888);

```

+ Options

		prod_id	branch_id	quantity
☐	Edit Copy Delete	231000000	1	800
☐	Edit Copy Delete	231000000	4	789
☐	Edit Copy Delete	231000469	2	567
☐	Edit Copy Delete	231003562	3	888
☐	Edit Copy Delete	236767676	5	444

☐ Check all With selected: Edit Copy Delete Export

☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

- Uses

```

1 INSERT INTO Uses VALUES
2 (234,50),
3 (1,30),
4 (334,20),
5 (234,75),
6 (23,10);

```

+ Options

		customer_id	discount_id
☐	Edit Copy Delete	23	10
☐	Edit Copy Delete	334	20
☐	Edit Copy Delete	1	30
☐	Edit Copy Delete	234	50
☐	Edit Copy Delete	234	75

☐ Check all With selected: Edit Copy Delete Export

☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

5. 5. Sub-Query:

This query retrieves the full name, email, and loyalty points of all customers who have at least 20 loyalty points.

Run SQL query/queries on database fall24_231000469:

```
1 SELECT CONCAT(F_name, ' ', Mid_name, ' ', L_name) AS full_name, email, loyalty_points
2 FROM Customers
3 WHERE loyalty_points >= 20;
```

Clear Format Get auto-saved query

☐ Bind parameters

Bookmark this SQL query:

[Delimiter:] ☒ Show this query here again ☒ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks

Hide query box

Showing rows 0 - 4 (5 total, Query took 0.0008 seconds.)

SELECT CONCAT(F_name, ' ', Mid_name, ' ', L_name) AS full_name, email, loyalty_points FROM Customers WHERE loyalty_points >= 20

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key: None

Options

	full_name	email	loyalty_points
☐ Edit Copy Delete	Maya F Karter	maKarter4@gmail.com	100
☐ Edit Copy Delete	Lewis H Marigold	leWis@gmail.com	78
☐ Edit Copy Delete	Selena V Gonzalez	selenagon@gmail.com	30
☐ Edit Copy Delete	Tamar K Jabar	tamy@gmail.com	39
☐ Edit Copy Delete	Winston L Schmidt	schmidt@gmail.com	24

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key: None

This query retrieves the payment ID, customer ID, amount, and payment date for all payments made using a specified method like a credit Card'.

Run SQL query/queries on database fall24_231000469:

1

2

3

```
SELECT payment_id, customer_id, amount, payment_date
FROM Payment
WHERE payment_method = 'Credit Card';
```

ClearFormatGet auto-saved query

☐ Bind parameters

Bookmark this SQL query:

[Delimiter :]

☒ Show this query here again

☒ Retain query box

☐ Rollback when finished

☒ Enable foreign key checks

Hide query box

Showing rows 0 - 0 (1 total, Query took 0.0011 seconds.)

```
SELECT payment_id, customer_id, amount, payment_date FROM Payment WHERE payment_method = 'Credit Card'
```

☐ Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

←→

▼

payment_idcustomer_idamountpayment_date

☐

Edit

Copy

Delete

100

234

199.99

2023-11-25

↑

☐ Check all

With selected:

Edit

Copy

Delete

Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table

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